

Software Documentation

1. Software Overview

Micropython is the logical centre of this entire arrangement, handling the actual operation of all computational logics and corresponding instructions for the hardware. The code simultaneously manages instructions which are most important in keeping the drone in stable flight at all times, especially with 3 completely different flight states meant to always be on standby for each other.

MIT App Inventor creates an easier interface for tracking/inspecting the game, specifically for users and larger spectator audiences. It can get quite intense so as to playing the game itself to be keeping the score but the app takes over, offering an overall better play experience.

2. System Logic

- 1. User switches drone on*
- 2. User taps on esp32 in bluetooth devices list*
- 3. The app screen shows "connected" (the esp32 and app are now connected via Bluetooth)*
- 4. User presses "play" button*
- 5. App sends confirmation to start game flight loop*
- 6. esp32 receives data and engages upward thrust in motors towards the apex of parabola.*
- 7. esp32 directs decreasing thrust toward one side, with ultrasound sensor ready for input.*
- 8. Screen shows graphical path of drone*
- 9. If ultrasound receives input - this is sent to esp32 to switch the direction of the parabola and continue the game*
- 10. Data sent to the app, "hit!" displayed on screen*
- 11. If no input is received, esp32 directs a timeout hover, and then resumes the game from the start.*
- 12. Data sent to app, "score to: " (player #) is displayed. Score count is also present.*
- 13. Score count stored in esp32.*
- 14. Once a "set" (first to 5 points) ends, the parabola loop restarts.*
- 15. Data sent to app: set count displayed in app.*

16. Set count calculated via score count and stored in esp32.
17. When game, set, match condition is fulfilled (2:1 sets to both players): data sent to app
18. App declares "winner"
19. Esp32 runs "timeout thrust" for drone

3. Flowchart

Visual representation of logic

https://miro.com/app/board/uXjVHbwkr9g=/?share_link_id=412391022789

4. App Flow (If MIT App Inventor)

https://miro.com/app/board/uXjVHbwkr9g=/?share_link_id=669102297637